Smart College and Course Finder

"Our idea is to develop a Smart College & Course Finder for Class 12 students that helps them choose suitable colleges and courses based on their academic stream, preferred course, location, and entrance-exam plans. The platform will provide personalised college recommendations along with admission eligibility, entrance-exam requirements, previous-year cutoffs, fees, and placement packages in one place, making the college-selection process simpler and less confusing." *Generated by LeapSpace, Sat Oct 10 2026*

Smart College & Course Finder for Class 12 Students: Quick Insights

Key Findings Table

Topic	Key finding	Implication for your platform	Citations
Personalization inputs	Effective recommenders combine academic profile variables with explicit preferences such as stream/branch, marks or prior performance, category, location, fees, and career goals rather than relying on one score alone.	Collect a structured student profile before recommending colleges or courses.	[1] [2] [3] [4] [5] [6] [7] [8]
Preference capture	Retrieved systems mostly gather preferences through forms, questionnaires, and explicit questioning; evidence for rich conversational preference elicitation is limited.	Start with simple forms/questionnaires; add chatbot interaction later only if it remains transparent.	[9] [10] [11] [12] [13]
Eligibility logic	A strong pattern is eligibility-first filtering: only colleges/courses satisfying constraints should remain before ranking.	Build a rule engine first, then a ranking engine.	[2] [11] [14] [15] [16] [17] [18] [19] [20]
Entrance-exam pathway	Some systems explicitly use exam scores or exam-related inputs such as JEE and CET for prediction and recommendation.	Include exam intent and score/rank fields as first-class inputs.	[21]

Topic	Key finding	Implication for your platform	Citations
Admission probability	Historical records, previous scores, and cutoff bands are commonly used to estimate admission chances or score proximity.	Use previous-year cutoff and score-band models to label options as ambitious, realistic, or safe.	[22] [23] [24] [25] [26]
Evaluation gap	Literature reports accuracy-like metrics more often than shortlist quality, calibration, or top-k usefulness.	Validate the product with ranking and decision-support metrics, not just prediction accuracy.	[22] [23] [24] [26] [27] [28] [29] [30]
Location	Location is a recurring factor, but the literature rarely specifies a clear formula for balancing geography against fees or admission probability.	Let users explicitly adjust location priority instead of hiding the trade-off.	[1] [8] [12] [14] [31]
Comparison data	Students benefit from fee, placement, rating/accreditation, faculty, facilities, and admission data, but few systems combine everything in one dashboard.	A unified comparison layer is a strong differentiator.	[8] [32] [33] [34] [35]
Explainability	Trust improves when recommendations are explained via feature rationales, graph/path reasoning, rule traces, or natural-language justifications.	Show why each college is recommended in plain language.	[36] [37] [38] [39] [40] [41] [42] [43]
Deployment constraints	Reliability depends on fresh, standardized, auditable data; known problems include incomplete records, cold start, fairness, privacy, and offline-online mismatch.	Prioritize data governance and update workflows from the start.	[34] [40] [44] [45] [46] [47] [48] [49] [50] [51] [52]

Direct Answer

Yes—your idea is well supported by the supplied research, but the strongest evidence points to a specific product architecture: an **eligibility-first, preference-aware recommender** that first collects stream, marks, category, preferred course, location, fee ceiling, and entrance-exam plans; then removes ineligible options using rules; then ranks the remaining colleges using cutoff proximity, admission likelihood, location fit, fees, and outcomes such as placements and institutional-quality signals [1] [14] [15] [19] [21] [32].

The best differentiation opportunity is **integration**. The evidence shows that existing systems often cover only part of the decision problem—such as comparison, cutoff prediction, or institutional quality—but rarely combine eligibility rules, previous-year cutoffs, fees, placements, and transparent explanations in one place [8] [33] [35] [36] [40].

For adoption, especially by students and parents, the platform should explain each recommendation in plain language—for example: “eligible through exam pathway,” “close to last year’s cutoff,” “within budget,” and “matches preferred city and course” [36] [41] [42] [53].

Confidence level: High - The evidence consistently supports eligibility-first filtering, structured student profiling, comparison features, and explainability as core design requirements across multiple sources [1] [2] [11] [14] [32] [36] [40].

Study Scope

This report synthesizes evidence from the supplied literature on educational recommendation, admissions decision support, college comparison, explainability, and deployment constraints, with emphasis on undergraduate and Class 12 decision support and Indian admissions relevance where available [1] [8] [14] [50]. The methods represented include rule-based and constraint-based systems, hybrid recommenders, collaborative filtering, boosting, regression, neural models, learning-to-rank approaches, multi-stage workflows, and explainable AI methods [11] [16] [27] [28] [29] [30] [36] [41].

Assumptions & Limitations

- The supplied evidence does **not** show a single validated production system that fully integrates stream, marks, category, course preference, location, fee ceiling, entrance-exam intent, previous-year cutoffs, fees, placements, and explainability in one Indian undergraduate platform [8] [33] [35].
- Reported evaluations are often narrow, emphasizing accuracy-like measures more than shortlist quality, calibration, top-k usefulness, or long-term student outcomes [22] [23] [24] [26].
- Evidence on the optimal weighting of location versus fees versus admission probability versus placements is limited, so any final ranking formula should be treated as a product choice to test empirically rather than a settled research result [8] [12].

Suggested Further Research

- Test ranking quality with shortlist-centric metrics such as top-k usefulness and decision satisfaction, since current studies underreport these outcomes [22] [23] [24] [26].
- Study transparent trade-off controls for location, affordability, and admission probability because existing work under-specifies this balancing step [8] [12].
- Investigate fairness across categories and socioeconomic groups in admissions recommendation workflows, given the governance concerns in the literature [40] [44].
- Evaluate dynamic policy-update pipelines for changing rules, thresholds, and cutoffs across years and exams [50] [51].
- Explore whether richer conversational elicitation improves student decisions over standard forms and questionnaires, since current evidence is limited [12] [13].

Full Report Body

Foundational Context

A Smart College & Course Finder sits at the intersection of two different decision-support tasks. The first is **deterministic eligibility checking**: can a student apply to a given college-course option under stream, subject, category, location, fee, and exam-pathway constraints? The second is **probabilistic ranking**: among eligible options, which are better fits or more attainable based on prior cutoffs, score bands, fees, outcomes, and preferences? This split is strongly aligned with the literature on constraint-based recommendation and predictive admissions modeling [15] [16] [24].

This distinction matters because the literature repeatedly shows that invalid recommendations undermine utility. Constraint-based recommendation is defined around retaining only items consistent with the active constraint set, while hybrid systems add learned ranking only after such filtering [15] [16] [17] [18]. In practical terms, a student-facing platform should not rank a program highly if the student is ineligible by stream, prerequisite, category rule, score threshold, or exam route [2] [11] [14] [19].

A second foundational point is that educational guidance is **multi-criteria**. The research does not support a one-number recommendation model. Instead, recommenders draw on profile data, explicit preferences, historical trends, and institutional attributes, then combine them in a structured way [1] [3] [4] [32]. Taken together, these findings suggest that your platform should be designed less like a generic search engine and more like a constrained decision-support system with transparent ranking logic [15] [36] [40].

Confidence level: High - The distinction between filtering and ranking appears consistently across the supplied recommendation and admissions sources [15] [16] [19] [24].

1. Personalized Student Profiling and College Matching

1.1 Student profile inputs

The literature strongly supports collecting a structured student profile that includes academic and preference variables. Across studies, recommender inputs include branch or stream, category or caste, location, fees, academic performance, and broader personal preferences or career goals [1] [2] [3] [4] [5] [6] [7] [8]. Some systems also incorporate entrance-exam scores or related test information such as JEE and CET for prediction or recommendation [21].

For your Class 12 use case, the evidence-backed minimum profile would include:

- academic stream or intended branch/course interest [1]
- marks or prior academic performance [3] [4] [7]
- category or caste where relevant to admissions rules [1]
- preferred location [1] [14]
- fee ceiling [1]
- entrance-exam intent and, if available, score or rank [21]

One supplied finding notes that academic information can also be transformed into an implicit rating matrix for collaborative filtering rather than only handled as form inputs [5]. That suggests a possible future recommendation layer, but the dominant pattern in the evidence remains explicit structured profiling rather than latent-only modeling [12] [13].

1.2 Preference elicitation methods

Preference elicitation in the retrieved literature is mostly straightforward. Systems gather preferences by asking direct questions, using questionnaires, or collecting explicit user input about interests and constraints [9] [10] [11] [12] [13]. The supplied evidence does not show strong use of interview-based, rating-task-based, or full conversational-agent elicitation for undergraduate recommenders; the closest described mode is explicit questioning [12] [13].

This has two practical implications. First, a form-driven onboarding flow is well supported and likely sufficient for an initial product [12] [13]. Second, if a chatbot is added, it should probably function as a user-friendly wrapper around explicit fields rather than as an opaque inference system, because the evidence base currently favors transparent structured collection [36] [40].

1.3 Eligibility-based recommendation logic

Constraint-based logic is one of the clearest themes in the evidence. A candidate should remain in the consideration set only if it satisfies the active constraints [15]. Related work also shows the value of prerequisite-aware recommendation and hybrid systems that combine rule-based filtering with learned ranking [16] [17] [18] [54] [55].

For your platform, that translates into a rule engine that checks:

- stream-to-course compatibility [11] [14]
- prerequisite subjects or course requirements [54] [55]
- category-linked reservation or cutoff conditions [2] [19]
- entrance-exam pathway validity [20] [21]
- location and fee constraints where the student treats them as hard filters [1]

Only after these checks should the system rank remaining options [16] [18]. Taken together, the research supports a two-layer architecture: a deterministic “Can I apply?” layer and a predictive “How suitable is this?” layer [15] [16] [24].

Confidence level: High - The evidence is strong and convergent on structured profiling plus eligibility-first filtering [1] [2] [11] [14] [15].

2. Admission Probability and Cutoff Estimation

2.1 Historical cutoff and score modeling

Several supplied studies estimate admission scores, score proximity, or cutoff-related outcomes from historical data. Examples include models based on high-school grades and school characteristics, regression and neural-network prediction from historical admission scores, and distance-to-previous-year-cutoff approaches for major choice [22] [23] [24]. Another study reports 93% accuracy for predicted cutoffs in classifying admissions status [25].

This supports including previous-year cutoff intelligence as a core product feature. A practical implementation grounded in the literature would label options by estimated attainability—for example, “safe,” “match,” or “ambitious”—based on a student’s profile relative to historical cutoffs or score bands [24] [25]. That would be more actionable than showing raw historic numbers alone [22] [23].

2.2 Ranking by fit or admission likelihood

Beyond simple eligibility, the literature shows broad use of ranking methods: collaborative filtering, fuzzy logic, graph models, boosting, and learning-to-rank style approaches are all used to order options by fit or expected success [11] [27] [28] [29] [30]. Some location-aware systems also combine rank, category, preferred branch, location, and fees in the recommendation process [1] [14].

For your platform, the evidence suggests ranking should be **multi-signal**, not cutoff-only. Suitable ranking signals include:

- admission likelihood or cutoff proximity [24] [25]
- course/branch preference match [1] [2]
- location preference match [12] [14]

- fee compatibility [1]
- institutional outcome or quality indicators where available [8] [32]

However, the literature does not provide a settled formula for weighting these against each other [8] [12]. This implies your ranking model should expose user-adjustable priorities or at least display the factors driving rank order [36] [40].

2.3 Metrics for recommendation quality

The evidence base is notably thinner on evaluation than on modeling. Reported metrics center on accuracy-like performance, and the supplied excerpts do not report precision@k, top-k hit rate, MAE, calibration, or user-acceptance metrics for many of the core admission-recommendation studies [22] [23] [24] [26]. One study reports 93% admissions classification accuracy, but that alone does not show whether the recommended shortlist is practically useful [25].

This is important for product design. A model can be statistically accurate yet still produce poor shortlist ordering or confusing user experiences. Inferred: the product should be evaluated not only on predictive correctness but also on shortlist quality, user trust, and decision usefulness. **Confidence level: Low** - This is an inference from the mismatch between the literature's metrics and the platform's actual decision-support goals [22] [23] [24] [26] [56] [57].

Confidence level: Medium - Prediction methods are well supported, but evidence on the best evaluation framework for your product is incomplete [22] [23] [24].

3. College Comparison: Fees, Placements, and Institutional Quality

3.1 Key comparison attributes

The supplied research shows that college decision-support tools often combine exam scores, ratings, placement scores, accreditation, tuition costs, geographic orientation, and alumni-related information to support recommendations [32]. User-facing systems may also include detailed institution data such as programs, faculty, admission requirements, and campus facilities [33] [34].

A strong comparison module for your platform should therefore display:

- fees or tuition [32] [33]
- placement indicators such as median salary or number of students placed where available [8] [32]
- ratings, accreditation, or quality signals [32] [35]
- faculty strength or institutional resources [8] [33]
- admission requirements and previous-year cutoffs [33] [35]

3.2 Data sources for comparison

The evidence highlights fragmented but valuable data sources. One personalized ranking system uses NIRF and JoSAA data for signals including median salary, number of students placed, faculty strength, and research funding [8]. Another admission-cutoff model combines student reviews, historical cutoff data, and NIRF scores [35]. A separate integrated information platform includes tuition fees, faculty profiles, admission requirements, campus facilities, and chatbot support, though not placement packages or accreditation signals in the abstract [33].

Taken together, these findings suggest a realistic data stack built from:

- official admissions and counseling data for eligibility and cutoffs [35]
- institutional-quality data sources such as ranking frameworks [8] [35]
- institution-level profile information for fees, faculty, and facilities [33]

3.3 Comparison workflow and shortlisting

The literature supports comparison, favorites, and shortlist-oriented workflows. Some systems explicitly provide college comparison tools, favorite lists, chatbots, and detailed institution views to help students choose [33] [34]. Broader guidance architectures also emphasize sequential processes of information collection, matching, recommendation, confirmation, and feedback [58].

For your product, the comparison layer should not be a passive table. It should support:

1. adding eligible colleges to a shortlist [33] [34]
2. sorting or filtering by user-defined criteria such as fees, location, or placements [8]
3. comparing a small set side by side on admissions, cost, and outcomes [32] [33]
4. revising recommendations when the student changes constraints [58]

Confidence level: High - The evidence is strong that integrated comparison features are useful, while fully unified dashboards remain rare and therefore product-differentiating [8] [32] [33] [34].

4. Explainability, Trust, and Deployment Constraints

4.1 Explainable recommendation styles

A recurring theme is that black-box recommendations reduce trust in high-stakes educational settings [36] [37] [38] [39] [40]. The supplied studies describe several explanation styles:

- feature-based rationales from course or item attributes [36]
- path or graph reasoning, including attention-weighted explanations [36] [39]
- proof or rule-trace explanations using formal reasoning representations [42]

- natural-language or LLM-generated justifications [41]

For your platform, the most practical style appears to be a hybrid explanation card combining rule traces and plain-language summary. For example: eligible by exam and category, close to historical cutoff, fits chosen city, and within fee range [36] [41] [42]. This matches the trust literature while remaining operationally simple [40].

4.2 User trust and usability

One platform aimed at students and parents emphasizes transparent, user-friendly presentation of personalized recommendations and admission-probability insights, even though the excerpt does not specify a particular explanation mechanism [53]. More broadly, transparent systems are repeatedly associated with stronger usability and adoption, while some decision-support systems also report high satisfaction or sensitivity transparency in evaluation [36] [56] [57].

This matters because your target users are likely making high-stakes decisions under uncertainty. The literature therefore supports:

- clear explanation of why a recommendation appears [36] [40]
- visible confidence or certainty cues around admission likelihood [53]
- simple comparison and workflow support rather than hidden model complexity [33] [34]

4.3 Practical constraints and mitigation

Operational constraints are a major theme. Existing systems depend heavily on the quality and completeness of historical admissions and profile data, making them vulnerable to sparse, inconsistent, or outdated records [45] [46] [47] [48] [49]. Reviews and system papers also identify cold-start issues, fairness concerns, privacy risks, limited interpretability, and mismatch between offline evaluation and live use [40] [44].

The supplied evidence suggests several mitigations:

- convert heterogeneous records into a unified database schema [46]
- use similarity-based retrieval when direct prediction is weak or data are sparse [47] [48]
- use predictive backfilling or ranking models such as regression, random forest, or CatBoost where appropriate [34] [49]
- maintain auditable workflows and update logic when rules or thresholds change [50] [51]
- support digitized sequential verification and administrative processes [52]

Taken together, these findings suggest that data governance is not a backend detail but a core product requirement [40] [44].

Confidence level: High - Trust, transparency, and data-governance constraints are among the most consistent themes in the supplied evidence [36] [40] [44] [51].

5. Recommended Product Architecture from the Evidence

A research-aligned architecture for your Smart College & Course Finder would include:

Layer 1: Student input and profiling

Capture stream, marks, category, preferred course, location, fee ceiling, and entrance-exam intent through structured forms or explicit question flows [1] [12] [13].

Layer 2: Eligibility engine

Apply deterministic rules for stream-course fit, prerequisites, category rules, cutoff or score thresholds, and exam pathways; discard invalid options before any ranking [15] [19] [20] [54].

Layer 3: Ranking engine

Score remaining options using admission likelihood, previous-year cutoff proximity, location fit, fee fit, and available institutional outcomes [1] [14] [24] [32].

Layer 4: Comparison dashboard

Display fees, placements or salary proxies, faculty/institutional quality, admission requirements, and shortlist comparison views [8] [33] [34].

Layer 5: Explanation layer

Generate a reason card for each recommendation using rule traces plus plain-language summaries [36] [41] [42].

Layer 6: Governance and updating

Track data source quality, refresh cutoffs and rules over time, and preserve auditability for changing admission criteria [40] [44] [50] [51].

Inferred: this layered design is likely to outperform an end-to-end opaque recommender for this use case because the admissions problem combines hard constraints with soft preferences. **Confidence level: Medium** - The architecture is a synthesis across multiple findings rather than a directly evaluated single system [15] [24] [36] [40].

6. Strategic Implications

Three practical conclusions follow from the evidence.

First, the product should compete on **decision clarity**, not only on prediction accuracy. The literature repeatedly points to fragmented tools and trust problems, creating room for a unified and explainable platform [32] [36] [40].

Second, **eligibility correctness** is foundational. A polished ranking system cannot compensate for recommending ineligible options [15] [19].

Third, **data freshness and provenance** are strategic assets. Because records, cutoffs, and admission pathways change, the operational value of the product will depend heavily on disciplined updating and transparent data quality labels [40] [44] [51].

Confidence level: High - These implications are strongly supported by recurring patterns in the evidence synthesis [15] [32] [40] [44].

Conclusion

The supplied research supports building your Smart College & Course Finder as a **two-stage, explainable admissions decision-support system**: collect structured student preferences and profile data; enforce eligibility first; then rank eligible colleges using historical cutoff signals, location, affordability, and outcome indicators [1] [14] [15] [24] [32]. The strongest product opportunity is to integrate data that are usually fragmented—eligibility rules, exam pathways, previous-year cutoffs, fees, placements, and institutional quality—while explaining every recommendation clearly to students and parents [8] [33] [35] [36] [40].

The main research gaps are not about whether such a platform is worthwhile, but about how best to weight competing criteria, how to evaluate shortlist quality in live use, and how to maintain fairness and data reliability over time [12] [26] [44] [51].

Overall confidence level: High - Core design choices are well grounded in the supplied evidence, though the optimal final ranking formula and live-impact evaluation remain under-specified [15] [24] [40] [44].

References

1. Kanjalkar, J.P., Patil, G.N., Patil, G.R., (...), Kanjalkar, P. (2023) Wise apply on a machine learning-based college recommendation data system. *Data-Driven Intelligent Business Sustainability*. https://doi.org/10.4018/979-8-3693-0049-7.ch018
2. Jaishree, R., Ida Christy, J.I., Abisha, D., (...), Abhirami, J. (2025) Smart College Admission Recommendation System. *Proceedings of 3rd International Conference on Sustainable Computing and Data Communication Systems, ICSCDS 2025*. https://doi.org/10.1109/ICSCDS65426.2025.11167729
3. AlNuaimi, A., Al Sinan, H., Alsuwaidi, R.M., Al Aghbari, Z. (2025) Recommender System for College Major Selection Based on Student Score. *4th International Conference on Applied Artificial Intelligence and Computing, ICAAIC 2025*. https://doi.org/10.1109/ICAAIC64647.2025.11331198

4. Delahoz-Domínguez, E.J., Hijón-Neira, R. (2024) Recommender System for University Degree Selection: A Socioeconomic and Standardised Test Data Approach. *Applied Sciences (Switzerland)*. https://doi.org/10.3390/app14188311
5. Vaidhehi, V., Suchithra, R. (2019) An enhanced approach using collaborative filtering for generating under graduate program recommendations. *2019 2nd International Conference on Advanced Computational and Communication Paradigms, ICACCP 2019*. https://doi.org/10.1109/ICACCP.2019.8882939
6. Nagaraj, P., Saiteja, K., Ram, K.K., (...), Muneeswaran, V. (2022) University Recommender System based on Student Profile using Feature Weighted Algorithm and KNN. *International Conference on Sustainable Computing and Data Communication Systems, ICSCDS 2022 - Proceedings*. https://doi.org/10.1109/ICSCDS53736.2022.9760852
7. Senthilselvi, A., Aakash, S., Hariharan, M.B., (...), Senthilpandi, S. (2024) GradExplore: A Holistic Approach to Assisting Higher Studies in Overseas Universities. *ICCDS 2024 - International Conference on Computing and Data Science*. https://doi.org/10.1109/ICCDS60734.2024.10560459
8. Goyal, A., Goyal, K., Pudi, V. (2025) Personalized Re-Ranking of Universities and Colleges Using NIRF and JoSAA Data. *2025 International Conference on Artificial Intelligence and Data Engineering, AIDE 2025 - Proceedings*. https://doi.org/10.1109/AIDE64228.2025.10987558
9. Khoja, Z., Shetty, S. (2018) Hybrid Recommender System for College Courses. *Proceedings - 2017 International Conference on Computational Science and Computational Intelligence, CSCI 2017*. https://doi.org/10.1109/CSCI.2017.203
10. Opranescu, V., Ioniţă, A.D. (2024) TOWARDS A RECOMMENDATION SYSTEM FOR AN EDUCATIONAL PROFILE IN SYSTEMS ENGINEERING. *UPB Scientific Bulletin, Series C: Electrical Engineering and Computer Science*. https://www.scopus.com/pages/publications/85194826862
11. Prasad, S.K., Sunderraman, R., Zhang, Y., Parvatiyar, A. (2004) A web-based game-oriented college selection system employing fuzzy rule trees. *Proceedings of the Hawaii International Conference on System Sciences*. https://www.scopus.com/pages/publications/12344336372
12. Kumar, K., Sinha, V., Sharma, A., (...), Vijay Krishna, B.S. (2022) AI-Assisted College Recommendation System. *Lecture Notes in Networks and Systems*. https://doi.org/10.1007/978-981-19-2894-9_11
13. Krishnasamy, T. (2021) A Novel study on Awareness and Preference of College Students towards Educational Based Mobile Application. *2021 7th International Conference on Advanced Computing and Communication Systems, ICACCS 2021*. https://doi.org/10.1109/ICACCS51430.2021.9441899
14. Tibile, H., Handigol, T., Kajagar, S., (...), Budihal, S.V. (2026) A Predictive Analytics Approach to College Recommendation Using XGBoost. *Lecture Notes in Networks and Systems*. https://doi.org/10.1007/978-3-032-06674-9_22
15. Uta, Mathias, Felfernig, Alexander, Le, Viet-Man, (...), Burgstaller, Tamim (2024) Knowledge-based recommender systems: overview and research directions. *Frontiers in Big Data*. https://doi.org/10.3389/fdata.2024.1304439

16. Zhu, Y., Lu, H., Qiu, P., (...), Niu, Z. (2020) Heterogeneous teaching evaluation network based offline course recommendation with graph learning and tensor factorization. *Neurocomputing*. https://doi.org/10.1016/j.neucom.2020.07.064

17. Lin, Y., Lin, F., Zeng, W., (...), Miao, C. (2022) Hierarchical reinforcement learning with dynamic recurrent mechanism for course recommendation. *Knowledge-Based Systems*. https://doi.org/10.1016/j.knosys.2022.108546

18. Esteban, A., Zafra, A., Romero, C. (2020) Helping university students to choose elective courses by using a hybrid multi-criteria recommendation system with genetic optimization. *Knowledge-Based Systems*. https://doi.org/10.1016/j.knosys.2019.105385

19. Arenas, A., Calsamiglia, C., Loviglio, A. (2021) What is at stake without high-stakes exams? Students' evaluation and admission to college at the time of COVID-19. *Economics of Education Review*. https://doi.org/10.1016/j.econedurev.2021.102143

20. Jiang, M., Li, J. (2023) When do stable matching mechanisms fail? The role of standardized tests in Chinese college admissions. *China Economic Review*. https://doi.org/10.1016/j.chieco.2023.101963

21. More, A., Patil, S., Savale, A., (...), Shinde, S. (2025) ML-Based College Predictor Telegram Bot Indian Engineering Colleges. *2025 1st International Conference on AIML-Applications for Engineering and Technology, ICAET 2025*. https://doi.org/10.1109/ICAET63349.2025.10932281

22. Arenas, A., Calsamiglia, C., Loviglio, A. (2021) What is at stake without high-stakes exams? Students' evaluation and admission to college at the time of COVID-19. *Economics of Education Review*. https://doi.org/10.1016/j.econedurev.2021.102143

23. Chen, Xiao, Peng, Yi, Gao, Yachun, Cai, Shimin (2022) A competition model for prediction of admission scores of colleges and universities in Chinese college entrance examination. *PLoS ONE*. https://doi.org/10.1371/journal.pone.0274221

24. Estevan, Fernanda, Gall, Thomas, Morin, Louis-Philippe (2018) Redistribution Without Distortion: Evidence from an Affirmative Action Programme at a Large Brazilian University. *The Economic Journal*. https://doi.org/10.1111/ecoj.12578

25. Shi, Y. (2020) Who benefits from selective education? Evidence from elite boarding school admissions. *Economics of Education Review*. https://doi.org/10.1016/j.econedurev.2019.07.001

26. https://www.sciencedirect.com/science/article/pii/S2661890725005770

27. Elbadrawy, A., Karypis, G. (2016) Domain-aware grade prediction and top-n course recommendation. *RecSys 2016 - Proceedings of the 10th ACM Conference on Recommender Systems*. https://doi.org/10.1145/2959100.2959133

28. Kalinin, A., Fizzano, P. (2008) A graph-based course recommender system. *23rd International Conference on Computers and Their Applications, CATA 2008*. https://www.scopus.com/pages/publications/84883368233

29. Kshirsagar, M.M., Gawande, U.H., Jain, S.R., (...), Girhepunje, V.G. (2025) A MULTI–MODEL MACHINE LEARNING FRAMEWORK FOR PERSONALISED COURSE RECOMMENDATION WITH DYNAMIC FEEDBACK. *African Journal of Applied Research*. https://doi.org/10.26437/emg0j607

30. Sapazhanov, Y., Batyrkhan, S., Togyssova, T., (...), Zhaksylyk, A. (2026) A Smart SIS Module for Course Planning: Curriculum-Aware Collaborative Filtering with Explainable Recommendation. *SIST 2026 - 2026 IEEE 6th International Conference on Smart Information Systems and Technologies, Conference Proceedings*. https://doi.org/10.1109/SIST61674.2026.11596355

31. López Turley, R.N. (2009) College proximity: Mapping access to opportunity. *Sociology of Education*. https://doi.org/10.1177/003804070908200202

32. Kumar, A., Kazoba, G.K., Kumar, A., (...), Gupta, A. (2026) An ML-driven decision support system for engineering college selection post-intermediate education in India. *Discover Global Society*. https://doi.org/10.1007/s44282-026-00351-4

33. Sahu, S.S., Kumar, R., Sahoo, S., (...), Mohanta, P. (2024) Machine Learning-Enabled Integrated Information Platform for Educational Universities. *Optimized Computational Intelligence Driven Decision-Making: Theory, Application and Challenges*. https://doi.org/10.1002/9781394242566.ch2

34. Sharma, V., Trehan, T., Chanana, R., Dawn, S. (2019) StudieMe: College Recommendation System. *2019 3rd International Conference on Recent Developments in Control, Automation and Power Engineering, RDCAPE 2019*. https://doi.org/10.1109/RDCAPE47089.2019.8979030

35. Girish, C., Umesh, T., Karthik, N., Vani, V. (2024) Anticipating Future College Admission Cutoffs: An Innovative Predictive Model Incorporating Student Reviews and Historical Admissions Cutoff Data Using Machine Learning. *IFIP Advances in Information and Communication Technology*. https://doi.org/10.1007/978-3-031-69982-5_13

36. Zabed Khan, M.A., Luo, D., Polyzou, A. (2025) Reasoning with Knowledge Graphs for Trustworthy Course Recommendation. *2025 IEEE 12th International Conference on Data Science and Advanced Analytics, DSAA 2025*. https://doi.org/10.1109/DSAA65442.2025.11248000

37. Chun, H.W., Ong, R.K., Khong, A.W.H. (2024) Reasonable Sense of Direction: Making Course Recommendations Understandable with LLMs. *Midwest Symposium on Circuits and Systems*. https://doi.org/10.1109/MWSCAS60917.2024.10658914

38. Kuflik, T., Dodge, J., Kleanthous, S., (...), Stumpf, S. (2022) TExSS 22: Transparency and Explanations in Smart Systems. *International Conference on Intelligent User Interfaces, Proceedings IUI*. https://doi.org/10.1145/3490100.3511165

39. Afreen, N., Balloccu, G., Boratto, L., (...), Marras, M. (2023) Towards Explainable Educational Recommendation through Path Reasoning Methods. *CEUR Workshop Proceedings*. https://www.scopus.com/pages/publications/85169154602

40. Nizam, T., Zafar, S., Biswas, S.S., (...), Alam, A.A. (2026) Towards Transparent Course Recommendations: The Role of XAI, Methodologies, and Emerging Directions. *Lecture Notes in Networks and Systems*. https://doi.org/10.1007/978-3-032-19672-9_9

41. Li, J., Lyu, Q., Qiu, W., Khong, A.W.H. (2025) Improving Course Recommendation Systems with Explainable AI: LLM-Based Frameworks and Evaluations. *Proceedings of the International Conference on Educational Data Mining*. https://doi.org/10.5281/zenodo.15870185

42. Narayanan, T., McGuinness, D.L. (2008) Towards leveraging inference Web to support intuitive explanations in recommender systems for automated career counseling. *Proceedings of the 1st International Conference on Advances in Computer-Human Interaction, ACHI 2008*. https://doi.org/10.1109/ACHI.2008.36

43. Takami, K., Dai, Y., Flanagan, B., Ogata, H. (2022) Educational Explainable Recommender Usage and its Effectiveness in High School Summer Vacation Assignment. *ACM International Conference Proceeding Series*. https://doi.org/10.1145/3506860.3506882

44. Mittal, S., Malhotra, I. (2026) Recent Advancements in Course Recommendation Systems in the Era of Large Language Models (2022-2025). *IEEE Access*. https://doi.org/10.1109/ACCESS.2026.3681211

45. Acharya, M.S., Armaan, A., Antony, A.S. (2019) A comparison of regression models for prediction of graduate admissions. *ICCIDS 2019 - 2nd International Conference on Computational Intelligence in Data Science, Proceedings*. https://doi.org/10.1109/ICCIDS.2019.8862140

46. Hasan, M., Ahmed, S., Abdullah, D.M., Rahman, M.S. (2016) Graduate school recommender system: Assisting admission seekers to apply for graduate studies in appropriate graduate schools. *2016 5th International Conference on Informatics, Electronics and Vision, ICIEV 2016*. https://doi.org/10.1109/ICIEV.2016.7760053

47. Sivasangari, A., Shivani, V., Bindhu, Y., (...), Vignesh, R. (2021) Prediction Probability of Getting an Admission into a University using Machine Learning. *Proceedings - 5th International Conference on Computing Methodologies and Communication, ICCMC 2021*. https://doi.org/10.1109/ICCMC51019.2021.9418279

48. Katti, J., Agarwal, J., Bharata, S., (...), Biradar, V. (2022) University Admission Prediction Using Google Vertex AI. *2022 1st International Conference on Artificial Intelligence Trends and Pattern Recognition, ICAITPR 2022*. https://doi.org/10.1109/ICAITPR51569.2022.9844176

49. Majeed Inamdar, A., Mhatre, T., Nadar, P., Joshi, S. (2022) Personalized College Recommender and Cutoff Predictor for Direct Second Year Engineering. *2022 IEEE 7th International conference for Convergence in Technology, I2CT 2022*. https://doi.org/10.1109/I2CT54291.2022.9825378

50. Trung, T.-N., Le, D.-N., Nguyen, T.-V., Nhu, G.-N. (2026) A forecast–optimization framework for data-driven university admissions. *Telematics and Informatics Reports*. https://doi.org/10.1016/j.teler.2026.100352

51. Traynor, Marian, Neill, Colin Mc, Roulston, Audrey (2022) Personal statement versus psychological test as admission to the nursing degree: an evaluation. *BMC Nursing*. https://doi.org/10.1186/s12912-022-00919-w

52. Lo, H.-W., Lin, S.-W. (2026) A hybrid Bayesian BWM-machine learning framework for university digital transformation assessment: Integrating expert clustering and predictive validation. *Technology in Society*. https://doi.org/10.1016/j.techsoc.2025.103170

53. Bhuvaneswaran, B., Vijay, P., Varshini, J., Yokeswaran, S. (2024) Equity-Enhancing AI: Integrating Counseling Marks for Fair Student Selection. *Proceedings of the 2024 10th International Conference on Communication and Signal Processing, ICCSP 2024*. https://doi.org/10.1109/ICCSP60870.2024.10543355

54. https://www.sciencedirect.com/science/article/pii/S1433305823001557

55. Tian, R., Cai, J., Li, C., Wang, J. (2024) Self-Supervised pre-training model based on Multi-view for MOOC Recommendation. *Expert Systems with Applications*. https://doi.org/10.1016/j.eswa.2024.124143

56. Oktavianti, E., Yudisti, I., Jiddan Wahono, A.M., (...), Suhandana, A.A. (2025) A Decision Support System for Tuition Fee Exemption Eligibility Using the Simple Multi Attribute Rating Technique. *Proceedings ICSINTESA 2025 - 2025 5th International Conference of Science and Information Technology in Smart Administration*. https://doi.org/10.1109/ICSINTESA68165.2025.11413756

57. de Los Santos, L.G., de Los Santos, G.N., de Los Santos, J.R.N., Villafuerte, R.S. (2024) SmartCourse: An Intelligent Recommender System for Course Offerings and Student Course Advising. *2024 7th International Conference on Vocational Education and Electrical Engineering: Charting the Course of Artificial Technology in Sustainable Society, ICVEE 2024*. https://doi.org/10.1109/ICVEE63912.2024.10823945

58. Tasatanattakool, P., Nongnuch, K., Wannapiroon, P., Nilsook, P. (2023) An Information Service Platform for Decision Support in Academic Admissions Using Data Fabrics and Artificial Intelligence. *International Journal of Interactive Mobile Technologies*. https://doi.org/10.3991/ijim.v17i21.41757